llmXive Automated Review of Role-Agent: Bootstrapping LLM Agents via Dual-Role Evolution

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided data in Table 1 and Table 2.

Major Discrepancies in Reported Metrics: In Section 4.2 ("Experimental Results"), the authors make several quantitative claims that do not align with the raw data presented in Table 1.

1. **Prompt-based Comparison:** The text states Role-Agent outperforms prompt-based methods (ReAct, Reflexion) by an "average of 78.0% on ALFWorld." Calculating the average of ReAct (12.8%) and Reflexion (21.8%) yields ~17.3%. The difference between Role-Agent (90.9%) and this average is 73.6 percentage points. A 78.0% figure does not correspond to either the absolute difference or the relative increase (~424%). This suggests a significant calculation error or a misinterpretation of the metric (e.g., perhaps comparing against a different baseline or a specific subset not clearly defined).
2. **Relative vs. Absolute Gains:** The text claims "relative gains of 4.2% / 6.9%" over GiGPO for ALFWorld/WebShop (Qwen-1.5B).

- ALFWorld: 90.9 - 86.7 = 4.2 (Correct as absolute points; relative is ~4.8%).
- WebShop: 71.9 - 65.0 = 6.9 (Correct as absolute points; relative is ~10.6%).

The text labels the WebShop gain as "6.9%" which implies a relative gain, but the number 6.9 is the absolute percentage point difference. This conflation is misleading, as a 6.9% relative gain would imply a score of ~69.5, not 71.9.

1. **Pick2 Task Claim:** The text cites a "+13.6% increase" on the Pick2 task (Qwen-7B). Table 1 shows 92.8% (Role-Agent) vs 79.2% (GiGPO). The difference is exactly 13.6 percentage points. However, the relative increase is (92.8-79.2)/79.2 17.2%. Using the "%" symbol without specifying "percentage points" creates ambiguity and technically misrepresents the magnitude of the relative improvement.

Recommendation: The authors must rigorously re-calculate all reported percentage improvements in Section 4.2. They should explicitly distinguish between "percentage point" differences (absolute) and "relative" percentage increases. The specific figure "78.0%" regarding prompt-based methods requires immediate verification and correction, as it currently appears factually unsupported by the table data.

Action items.

- [**writing**] The claim in Section 4.2 that Role-Agent outperforms prompt-based methods by '78.0% on ALFWorld' is mathematically inconsistent with Table 1. The average success rate for Role-Agent (90.9%) vs. the average of ReAct/Reflexion (~17.3%) represents a relative increase of ~424%, or an absolute increase of ~73.6 percentage points. The '78.0%' figure appears to be a calculation error or a mislabeled metric that requires correction to accurately reflect the data.
- [**writing**] The claim in Section 4.2 that Role-Agent outperforms GiGPO with 'relative gains

of 4.2% / 6.9%' on ALFWorld/WebShop (Qwen-1.5B) is inconsistent with Table 1. The data shows ALFWorld gains of 4.2% (90.9 vs 86.7) but WebShop gains of 6.9% (71.9 vs 65.0) are actually 6.9 percentage points, which is a ~10.6% relative gain. The text conflates absolute percentage point differences with relative percentage gains, potentially overstating the improvement on WebShop.

- [**writing**] The claim in Section 4.2 that Role-Agent shows a '+13.6% increase on the Pick2 task' (Qwen-7B) is ambiguous and likely inaccurate. Table 1 shows Pick2 scores of 92.8% (Role-Agent) vs 79.2% (GiGPO). This is an absolute difference of 13.6 percentage points, but a relative increase of ~17.2%. The text should explicitly state '13.6 percentage points' to avoid confusion with relative improvement metrics.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear schematic diagram that effectively illustrates the three distinct approaches described in the caption. The visual elements (lego figures, environments, and flow arrows) are distinct and legible, and the text bubbles provide immediate context for the limitations of the first two methods versus the proposed solution.

Figure 2

Figure 2 provides a clear and comprehensive visual overview of the Role-Agent framework, effectively illustrating the dual roles of the LLM as both agent and environment. The diagram logically maps the flow from state prediction and reward calculation to failure mode analysis and data distribution reshaping, aligning perfectly with the provided caption.

Figure 3

The figure effectively displays the running dynamics with clear legends and error bars, but the caption contains a formatting artifact ('black') and the right y-axis label could be more precise regarding the metric definition.

Figure 4

The figure effectively visualizes the accumulation of failure modes over training steps, but the legend contains entries that do not appear in the stacked bars, and the legend layout is dense.

Figure 5

Figure 5 effectively illustrates the case study described in the caption, clearly showing the progression from a failed trajectory to failure mode analysis and task retrieval. The visual layout is uncluttered, and the text within the panels is legible and consistent with the figure's stated purpose.

Figure 6

The figure presents a time breakdown with a confusing y-axis break and a 'Total' bar that does not mathematically correspond to the sum of its components, undermining the data's validity.
Action items.

- [**writing**] Figure 3: The caption contains the artifact 'black' at the beginning, which appears to be a formatting error or leftover LaTeX command.
- [**writing**] Figure 3: The y-axis label on the right plot ('Rollout Probs Diff Mean') is ambiguous; the caption describes it as 'averaged difference', but the label does not explicitly state the units or the specific metric being averaged.
- [**science**] Figure 4: The legend lists 12 failure modes, but the stacked bars only display 10 distinct colors; 'Exhaustive Exploration Failure' and 'Action Format Error' are missing from the visual data.
- [**writing**] Figure 4: The legend is cluttered with 12 items in a dense 3-column layout, making it difficult to map specific colors to failure modes.
- [**science**] Figure 6: The 'Total' bar (519.92) is not the sum of the preceding components (approx. 350s), and the 'Rollout' bar (175.36 + 18.63) does not align with the 'Total' bar's magnitude, making the breakdown mathematically inconsistent and misleading.
- [**writing**] Figure 6: The y-axis contains a break (indicated by the double slash) but lacks a clear visual gap or distinct scale change, making the transition between 30 and 200 visually confusing.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized acronyms and coined terms that are not defined at their first occurrence, creating a barrier for non-specialist readers. Specifically, the term "Agentic Reinforcement Learning (ARL)" appears in the Introduction without expansion. Similarly, the core components "World-In-Agent (WIA)" and "Agent-In-World (AIW)" are introduced as acronyms immediately, forcing the reader to memorize these labels before understanding the mechanisms they represent.

In the Methodology section, the metric "Longest Matching Subsequence (LMS)" is introduced without definition, assuming the reader is familiar with this specific string-matching technique in the context of RL rewards. Additionally, phrases like "bootstrapped co-evolution" and "state-grouped advantage" are used frequently without plain-English paraphrasing. While these terms are precise for experts, the paper would benefit from replacing them with more descriptive language (e.g., "self-improving loop" or "grouping actions by identical states") at least once to ensure clarity for a broader audience. The abstract and introduction are particularly dense with these undefined terms, which should be expanded or simplified to meet the standard of accessibility.

Action items.

- [**writing**] The manuscript relies heavily on specialized acronyms and coined terms that are not defined at their first occurrence, creating a barrier for non-specialist readers. Specifically, the term "Agentic Reinforcement Learning (ARL)" appears in the Introduction without expansion. Similarly, the core components "World-In-Agent (WIA)" and "Agent-In-World (AIW)" are introduced as acronyms immediately, forcing the reader to memorize these labels before understanding the mechanisms they represent. In the M

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally sound in its high-level narrative, but there are specific disconnects between the claimed mechanisms and the mathematical or procedural descriptions provided.

First, the central claim of "bootstrapped co-evolution" (Introduction, Section 1) implies a dynamic feedback loop where the environment adapts to the agent's current state in real-time. However, the description of the Agent-In-World (AIW) module (Section 3.2) reveals that the "environment" merely retrieves tasks from a pre-existing, static library of historical failures. The data distribution is adjusted by re-weighting samples from this fixed set, not by generating new, adaptive challenges or modifying the environment's rules. This creates a logical gap: the system performs *curriculum learning* based on past errors, not true *co-evolution* where the environment itself evolves. The terminology "co-evolution" is therefore an overstatement of the mechanism described.

Second, the justification for the reward modulation formula in the World-In-Agent (WIA) module (Section 3.1) contains a subtle logical flaw. The authors state that using multiplication prevents failed trajectories from being rewarded solely for plausible state predictions. While mathematically true (0 * x = 0), the text frames this as a "reliability-aware modulation." If the task reward is zero, the predictive reward has no effect whatsoever; it does not "weaken" a non-existent advantage, it simply vanishes. The logical link between the formula and the claim of "modulating" reliability in failed cases is weak, as the mechanism is a hard gate rather than a soft modulation.

Finally, the explanation for the performance drop with large prediction horizons H (Section 4.3) cites "reward hacking" via "speculative guesswork." Given that the predictive reward is strictly based on the Longest Matching Subsequence (LMS) against *ground-truth* states, an agent cannot "hack" the reward by guessing a state that matches its own hallucination unless that hallucination coincidentally matches the ground truth. The paper does not provide a logical mechanism for how speculative guessing would increase the LMS score against the ground truth, making the "reward hacking" claim unsupported by the described metric.

Action items.

- [**science**] The claim of "co-evolution" contradicts the static offline library used in AIW (Sec 3.2). The environment re-weights existing tasks rather than generating new adaptive challenges, making "co-evolution" an overstatement of the mechanism.
- [**writing**] The WIA reward formula R__t = R__task * (1 + R__pre) is described as "modulation," but if R__task is 0, the term vanishes entirely. This is a hard gate, not a soft modulation of reliability for failed trajectories as claimed.
- [**science**] The claim that large H causes "reward hacking" via "speculative guesswork" is unsupported. Since R__pre relies on LMS against ground truth, hallucinations cannot increase the score unless they match reality, which is not hacking.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims regarding the autonomy and generality of the "bootstrapped co-evolution" that extend beyond the provided evidence.

First, the Introduction and Contribution 1 assert that the method achieves "bootstrapped agent-environment co-evolution without human supervision." However, the "Agent-In-World" (AIW) component relies on a fixed, pre-defined taxonomy of failure modes (listed in the Appendix table, e.g., "wrong_target_location", "missing_precondition"). The LLM is prompted to map failures to these specific categories rather than discovering novel failure modes autonomously. This reliance on a human-curated schema contradicts the claim of being entirely "without human supervision" in the evolution process. The text should be tempered to reflect that the *selection* of tasks is automated, but the *definition* of failure modes is supervised.

Second, the claim that the framework endows agents with "substantial generalization capabilities" (Introduction, Conclusion) is partially undermined by the Search-Augmented QA results in Table 2. On the NQ dataset (an in-domain test set), Role-Agent (40.1%) underperforms the GiGPO baseline (42.0%). The authors dismiss this by attributing it to "stronger generalization capabilities rather than overfitting," which is a non-sequitur; typically, lower in-domain performance suggests a lack of specialization or a trade-off, not superior generalization. This specific interpretation is an overreach and should be revised to honestly acknowledge the performance dip on NQ.

Finally, there is a discrepancy in the efficiency claims. The "Efficiency Study" section states the method induces "only about 5.2% extra computation," while the "Computational Overhead" appendix reports a total throughput drop from 150 to 109 tokens/sec, which is a ~27% increase in latency. The 5.2% figure appears to isolate the AIW feedback cost, ignoring the significant cost of the predictive rollouts (WIA). This selective reporting overstates the efficiency of the full system.

Action items.

- [**science**] The claim of 'bootstrapped co-evolution' without human supervision (Introduction, Contribution 1) is overstated. The failure modes (Table in Appendix A) are pre-defined categories (e.g., 'wrong_target_location'), and the retrieval logic relies on these fixed labels rather than emergent, unsupervised discovery. The paper should clarify that the 'evolution' is guided by a fixed taxonomy, not purely bootstrapped.
- [**writing**] The statement that Role-Agent achieves 'substantial improvements' and 'generalization capabilities' (Introduction, Conclusion) is not fully supported by the search-QA results (Table 2). Role-Agent underperforms GiGPO on the NQ dataset (40.1% vs 42.0%). The authors attribute this to 'stronger generalization' rather than overfitting, but this contradicts the standard interpretation of lower in-domain performance. This specific claim needs retraction or stronger evidence.
- [**writing**] The efficiency claim of 'only about 5.2% extra computation' (Efficiency Study) conflicts with the 'Computational Overhead' appendix which reports a ~27% total overhead (109 tokens/sec vs 150 tokens/sec). The 5.2% figure likely refers only to the AIW feedback loop, excluding the predictive rollout cost. The text should be precise to avoid misleading readers about the total inference cost.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a novel framework for bootstrapping LLM agents via dual-role evolution, where a single model acts as both agent and environment. From a safety and ethics perspective, the work is generally sound as it operates within controlled, text-based simulation environments (ALFWorld, WebShop, QA datasets) rather than real-world physical or high-stakes digital systems. However, several areas require clarification to ensure full transparency and address potential risks inherent in self-evolving systems.

First, the "The Use of Large Language Models" section in the Appendix (lines 1045-1050) states that LLMs were used for grammar and coding assistance. Given that the core methodology (AIW) relies heavily on the LLM to generate failure mode analyses and retrieve tasks, the authors should explicitly clarify the extent of LLM involvement in generating the *data* and *results* versus just the *text*. It is crucial to distinguish between human-designed prompts and LLM-generated content to ensure reproducibility and prevent the misattribution of scientific claims to the model itself.

Second, the "Agent-In-World" (AIW) component creates a closed loop where the agent analyzes its own failures to reshape the training distribution. While the paper discusses performance gains, it lacks a discussion on the safety of this self-reinforcing loop. There is a risk of "reward hacking" or the amplification of biases, where the model might learn to exploit the failure-analysis prompt to generate high rewards without genuinely improving task performance, or converge on a narrow set of failure modes that reinforce specific biases. A brief discussion on potential failure modes of the co-evolution loop itself, and any safeguards implemented, would strengthen the ethical robustness of the claim.

Finally, the use of the WebShop benchmark, which simulates a large-scale e-commerce environment, raises minor data privacy considerations. Although the environment is simulated, the agent interacts with product data and potentially generates search queries or actions that could mimic real-world user behavior. The authors should include a statement confirming that no real user data was used or exposed and that the simulation adheres to standard data privacy protocols, even in a synthetic setting.

Overall, the paper does not present immediate dual-use risks or severe ethical violations, but the transparency regarding LLM-generated content and the safety of the self-evolution loop requires minor revisions.

Action items.

- **[writing]** The 'The Use of Large Language Models' section (Appendix) states LLMs were used for grammar and coding assistance but does not explicitly confirm that the LLMs were not used to generate the failure mode analyses, trajectory data, or the core experimental results. Given the method's reliance on LLM-generated feedback, a clearer distinction between human-designed prompts and LLM-generated content is required to ensure reproducibility and transparency.
- **[writing]** The paper describes a self-reinforcing loop where the LLM analyzes its own failures to curate future training data (AIW). There is no discussion of safeguards against 'reward hacking' or the amplification of specific biases (e.g., the model learning to exploit the failure-mode analysis prompt rather than solving the task). A brief discussion on potential failure modes of the co-evolution loop itself is needed.

- **[writing]** The WebShop benchmark involves simulating interactions with a large-scale e-commerce environment (1.18M products). The paper does not address data privacy or the potential for the agent to inadvertently generate or expose sensitive user data (e.g., credit card numbers, addresses) during the 'click' or 'search' actions, even in a simulated setting. A statement on data safety protocols is recommended.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling framework for agent-environment co-evolution, but the scientific evidence supporting the robustness of the claims requires strengthening in three key areas.

First, the ablation study in Table 3 (lines 330-340) reports performance drops for the -w/o AIW and -w/o Predictive Reward variants but lacks measures of statistical significance. While Appendix B (Table 4) provides standard deviations for the main baselines (GRPO, GiGPO, Role-Agent), it omits the ablated variants. Given the relatively small sample size of benchmarks (e.g., ALFWorld has a limited number of test episodes), a 5.0% drop on WebShop could potentially be within the noise floor of the training process. The authors should re-run the ablation experiments with at least 3-5 random seeds and report the mean and standard deviation for these specific variants to confirm the gains are reproducible.

Second, the validity of the World-In-Agent (WIA) component hinges on the Longest Matching Subsequence (LMS) metric used to compare predicted and actual states (Eq. 4, line 135). The paper reports a point-biserial correlation of 0.41 between predictive reward and outcome reward (Appendix B, line 465). However, LMS is a strict string-matching metric that may fail to capture semantic equivalence. In text-based environments, an agent might predict a state that is functionally identical but phrased differently (e.g., "the door is open" vs. "you see an open door"). If the LMS metric penalizes these valid predictions, the reward signal becomes noisy, potentially hindering learning. The authors should provide an analysis of the false-negative rate of the LMS metric or demonstrate that the state representations in their chosen benchmarks are sufficiently canonical to make LMS a reliable proxy for semantic correctness.

Third, the Agent-In-World (AIW) component relies on an LLM to retrieve tasks based on failure modes. The paper claims this creates a "targeted" data distribution but does not quantify the quality of this retrieval. If the LLM retrieves tasks that are only superficially similar but do not address the root cause of the failure, the training signal could be diluted or even harmful. The authors should include a quantitative evaluation of the retrieval process, such as the precision of retrieved tasks against a ground-truth set of relevant tasks, or a human evaluation of the failure mode analysis and retrieval steps (as illustrated in Figure 6). Without this, the claim that AIW effectively "reshapes the training data distribution" remains largely anecdotal.

Action items.

- [**science**] The ablation study in Table 3 lacks standard deviations for the -w/o AIW and -w/o Predictive Reward variants. Re-run these ablations with multiple seeds to confirm the reported gains are statistically significant and not due to random variance.
- [**science**] The predictive reward relies on LMS string matching, which may penalize semantically correct but textually different states. Provide evidence that LMS is robust to semantic equivalence in these benchmarks or analyze the false-negative rate of the state prediction metric.
- [**science**] The AIW retrieval mechanism lacks quantitative validation. Report precision/recall metrics for the retrieved tasks or provide a human evaluation of the failure mode analysis to prove the data distribution shift is actually targeting the identified deficiencies.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the paper is generally sound in its application of standard RL metrics (success rate, score) and ablation studies. However, several areas require stronger statistical rigor to fully support the claims of superiority and stability.

First, the reporting of variance is inconsistent. While Table 3 in Appendix B correctly reports mean ± standard deviation over three runs for the proposed method and key baselines, the main results in Table 1 and Table 2 only present point estimates (means). Without standard errors or confidence intervals for the baselines, it is impossible to determine if the reported improvements (e.g., the ~4% gain over GiGPO) are statistically significant or within the noise of the training process. The authors should re-run baselines for at least three seeds and report the variance in the main tables to enable proper significance testing.

Second, the correlation analysis in Appendix B (Section "Relation between predictive reward and outcome reward") reports a point-biserial correlation of 0.41 with $p<0.01$ based on $N=200$ rollouts. While the p-value suggests significance, the sample size is relatively small for establishing a robust relationship in stochastic RL environments. The authors should report the 95% confidence interval for this correlation coefficient. Furthermore, it is unclear if this correlation was computed on a single seed or averaged across multiple seeds; reproducibility of this specific statistical finding requires clarification on the experimental setup.

Third, the state grouping mechanism relies on a fixed Longest Matching Subsequence (LMS) similarity threshold of 0.9. The paper states this is kept from GiGPO for "fair comparison," but this does not constitute a statistical justification for the threshold's optimality in the current context. A sensitivity analysis (similar to the one provided for hyperparameter α and H) or a statistical test showing that 0.9 is the optimal cutoff for minimizing state conflation while maximizing grouping efficiency would strengthen the methodological validity.

Finally, the ablation study in Table 3 shows performance drops when components are removed, but without error bars, the magnitude of these drops (e.g., 5.0% on WebShop) cannot be statistically validated. Including standard deviations for the ablated variants is necessary to confirm that the observed drops are significant.

Action items.

- [**writing**] The statistical analysis in the paper is generally sound in its application of standard RL metrics (success rate, score) and ablation studies. However, several areas require stronger statistical rigor to fully support the claims of superiority and stability. First, the reporting of variance is inconsistent. While Table 3 in Appendix B correctly reports mean ± standard deviation over three runs for the proposed method and key baselines, the main results in Table 1 and Table 2 only present point e

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and generally well-structured argument for the Role-Agent framework. The writing is mostly fluent, and the logical flow from the problem statement to the proposed solution and experimental validation is coherent. The abstract effectively summarizes the core contributions, and the introduction successfully motivates the need for agent-environment co-evolution.

However, there are several specific instances where sentence-level grammar, punctuation, and phrasing impede clarity or violate standard academic conventions. In the Introduction, a semicolon is incorrectly used to join two independent clauses where the second begins with a capital letter ("...rewards; The environment fails..."), creating a grammatical error. In Section 3.1, the transition to the GRPO equation uses the non-idiomatic phrase "as the following" instead of "as follows," and the subsequent variable definitions suffer from inconsistent spacing around mathematical symbols, which affects readability.

Furthermore, in Section 3.2, the sentence structure regarding the "State Grouping" methodology contains a dangling modifier ("Instead of employing..., following GiGPO..."), which momentarily confuses the subject of the action. Finally, in the results discussion (Section 4.2), the claim of a "78.0%" improvement lacks precision regarding whether this is an absolute or relative gain, which is critical for interpreting the magnitude of the results. Addressing these specific grammatical and stylistic issues will significantly enhance the professional quality and readability of the paper.

Action items.

- [**writing**] In the Introduction, fix the semicolon splice: '...rewards; The environment fails...' should use a period or lowercase 'the' to correct the grammar.

- **[writing]** In Section 3.1, change 'as the following' to 'as follows' and fix inconsistent spacing around math symbols (e.g., '1± ') to improve readability.
- **[writing]** In Section 3.2, rephrase the dangling modifier 'Instead of employing..., following GiGPO...' to clearly attribute the action to the authors.
- **[writing]** In Section 4.2, clarify if the '78.0%' improvement is an absolute percentage point gain or a relative increase to prevent misinterpretation.